

Consistent System :-

- ↳ Given a set of logical statements, they should maintain consistency.
- ↳ These set of statements should not contain any contradictions.

eg. A : It is sunny or it is warm.

B : It is sunny.

C : It is not warm.

$P \rightarrow$ It is sunny.

$Q \rightarrow$ It is warm.

$$A \equiv P \vee Q$$

$$B \equiv P \rightarrow$$

$$C \equiv \neg Q$$

$\checkmark$ B		$\checkmark$ C	$\checkmark$ A	
$\odot P$	Q	$\odot \neg Q$	$\odot P \vee Q$	
T	F	T	T	① $\checkmark$
T	T	$\boxed{F}$	T	② $\times$
$\boxed{F}$	T	$\boxed{F}$	T	③ $\times$
$\boxed{F}$	F	T	$\boxed{F}$	④ $\times$

eg. A: If it is snowing, then it is cold.

B: It is snowing.

C: It is not cold.

P: It is snowing.

Q: It is cold.

A $\equiv$ $P \rightarrow Q$

B $\equiv$ P (T)

C $\equiv$ $\neg Q$ (T)

B		C		A
(P)	Q	($\neg Q$)	($P \rightarrow Q$)	X
T	F	T	<u>F</u>	

(H.W.) Check the consistency of the following statements:-

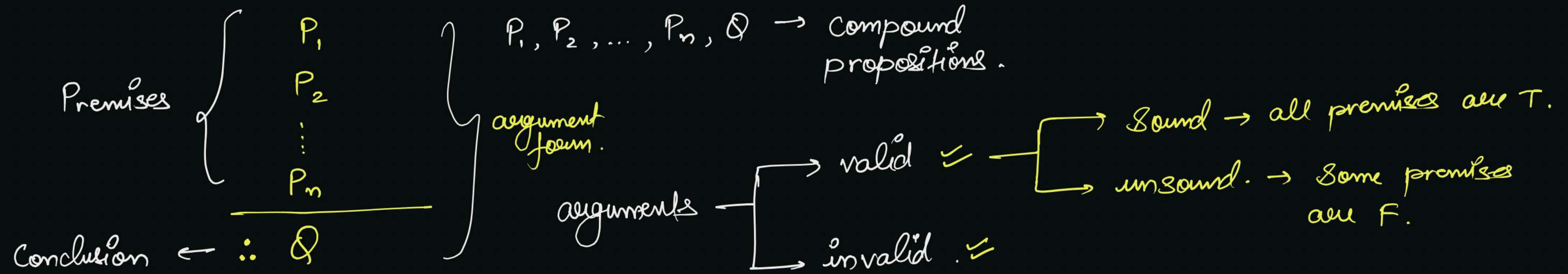
(1) A: If it is raining, then the ground is wet.
B: It is raining or it is snowing.
C: The ground is not wet.

✓

(2) A: If it is sunny, then it is warm.
B: It is sunny or it is snowing.
C: It is not warm.

X

Argument :- A set of statements designed to establish the truth of a particular conclusion.



- ↳ An argument is valid, if the conclusion logically follows the premises.
- ↳ In a valid argument, if the premises are true, then the conclusion must be true.

- ↳ The argument form is called valid, provided

$$\underbrace{[P_1 \wedge P_2 \wedge \dots \wedge P_n]}_X \rightarrow Q \text{ is a tautology.}$$

X	Q	$X \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Sound valid arguments: (T, T) and (T, F)

unsound valid arguments: (F, T) and (F, F)

eg. P_1 : If you have a current password, then you can log onto the network.

P_2 : You have the current password.

$\therefore Q$: You can log onto the network.

P : You have the current password.

Q : You can log onto the network.

P_1 : $P \rightarrow Q$

P_2 : P

Q : Q

$((P \rightarrow Q) \wedge P) \rightarrow Q$
is a tautology?

$\overline{P}$	$\overline{Q}$	$\overline{x}$ $P \rightarrow Q$	$\overline{y}$ $x \wedge P$	$y \rightarrow Q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

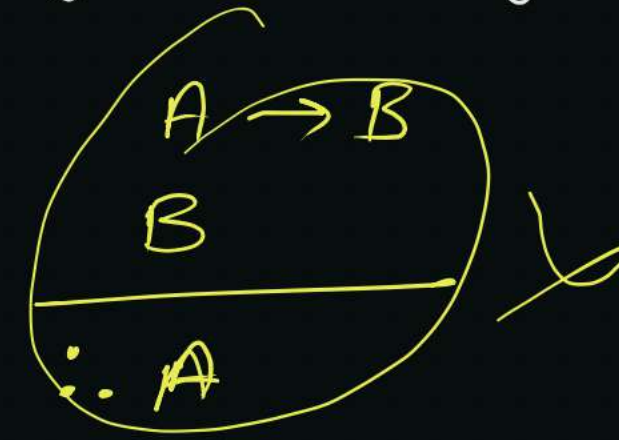
$$\frac{A \rightarrow B \quad A}{\therefore B} \quad \checkmark$$

Tautology $\rightarrow \therefore$ It's a valid argument.

Q. P_1 : If you have a current password, then you can log onto the network.

P_2 : You can log onto the network.

$\therefore Q$: You have the current password.



p : You have the current password.

q : You can log onto the network.

P_1 : $p \rightarrow q$

P_2 : q

Q : p

$((p \rightarrow q) \wedge q) \rightarrow p$
is a tautology?

③

$P_1 \wedge P_2 \dots \wedge P_m \rightarrow Q$

$\boxed{p}$	$\overline{q}$	$\overline{(x)}$ $p \rightarrow q$	$\boxed{y}$ $(x \wedge q)$	$y \rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	$\boxed{F}$
F	F	T	F	T

not a tautology
 $\therefore$ Invalid arguments.

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